

Roshna Roy

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EDUCATION

Jönköping University

Master's in Artificial Intelligence

Sweden

2024 – 2026

MG University

M.Sc. Computer Science

India

2018 – 2020

MG University

Bachelor of Computer Applications

India

2015 – 2018

TECHNICAL SKILLS

Programming Languages: Python, C++ , Java, PHP, HTML

Libraries and Tools: PyTorch, Scikit-learn, TensorFlow, Keras, Pandas, NumPy, KNIME, Flask

AI and Machine Learning: LLM, RAG, CoT, Supervised and Unsupervised Learning, Deep Learning (CNNs, RNNs), Natural Language Processing (NLP), Computer Vision

WORK EXPERIENCE

Master Thesis – Artificial Intelligence

Volvo Cars, Gothenburg, Sweden

Jan 2026 - Jun 2026

- Developed an AI-driven framework for context-aware root cause analysis of Android Automotive (AAOS) logs using LLMs and RAG.
- Built structured log analysis pipelines to extract, correlate, and explain failure patterns from large-scale unstructured logs.
- Evaluated and optimized the framework on public and industrial datasets, improving diagnostic accuracy and explainability.

Machine Learning Intern

Husqvarna Sverige, Jönköping, Sweden

Nov 2025 – Dec 2025

- Designed and developed ontology-driven semantic scene graphs for autonomous mower environments to support AI training data generation.
- Created structured knowledge representations and object relationship models to generate semantically consistent synthetic datasets.
- Integrated scene graphs with generative AI models to improve the realism, explainability, and validation of computer vision training data.

Software Developer

BETA, Kerala, India

Aug 2022 – Jul 2024

- Designed and maintained Python-based backend applications in production environments.
- Optimized application performance through debugging and SQL query tuning.
- Participated in code reviews, testing, and deployment processes.

Subject Matter Expert – Computer Science

Chegg India, Kerala, India

Sep 2021 – Oct 2022

- Provided academic support in programming, algorithms, data structures, and databases.
- Solved complex technical problems while maintaining high quality and accuracy standards.

PROJECTS

- **DLFashionInspired – Sustainable Fashion Classification**, Designed lightweight CNN and ensemble learning model and that achieved 94 percentage accuracy on DeepFashion dataset and 50 percentage on SLOWFASHION dataset.
- **House Price Prediction Using Machine Learning**, Developed predictive models using Decision Tree, Random Forest, and XGBoost and applied feature engineering and data augmentation to achieve high accuracy.
- **Analysis of Shipment Costs and Carrier Performance (PLAR)**, Applied CRISP-DM methodology on 60,000+ shipments using KNIME and built predictive models and dashboards to optimize logistics costs and carrier efficiency.

CERTIFICATIONS

- Python for Data Science and Machine Learning Bootcamp (Udemy)
- Certificate in Word Processing and Data Entry (Rutronix)
- CompTIA Server+ Certification Exam Training SK0-005 Server Plus Server Hardware Installation Administration Management(Udemy)